

BYOD: BUILD YOUR OWN DLM - EFFICIENT AUTOREGRESSIVE-TO-DIFFUSION LANGUAGE MODEL CONVERSION

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ABSTRACT

Diffusion language models (DLMs) permit parallel token updates and flexible generation orders, but leading systems rely on costly pretraining or continued full-model training. We ask how efficiently diffusion generation can instead be induced in existing instruction-tuned autoregressive models. Build Your Own DLM (BYOD) replaces causal attention with bidirectional attention and trains only query/value LoRA adapters under a masked denoising objective, leaving 94.2–95.9% of parameters frozen. Without architecture-specific tuning, the same recipe converts Gemma 2 9B, Llama 3.1 8B, Qwen2.5 7B, and Ministral 8B, with each 25,000-update conversion processing at most 102.4M token positions in 3.1–4.0 hours on one Google Colab GPU. All four models generate from fully masked sequences and retain substantial parent-model knowledge. BYOD-Gemma is the strongest conversion on our nine-task evaluation, reaching 53.0% average accuracy compared with 51.9% for LLaDA 8B Instruct in the same harness. Fluency remains the principal limitation: at 64 NFE, converted-model perplexity is 8.3–11.1 versus 4.1 for LLaDA and 2.2–3.0 for the AR parents, with generally lower Distinct- n scores as well. BYOD therefore establishes a practical, reproducible route for broad experimentation with diffusion generation while identifying the quality gap that efficient conversion must next close.

1 INTRODUCTION

Autoregressive (AR) language models generate a sequence from left to right. This method has proven extremely effective due to its simple learning target and effective scaling capabilities. However, it also has trade-offs: it fixes the generation order, limits the model to unidirectional attention, and predicts only one new token at a time. Diffusion language models (DLMs) instead begin with a corrupted sequence and can iteratively reconstruct multiple positions at a time using bidirectional context (Austin et al., 2021a; Sahoo et al., 2024; Shi et al., 2024). Their number of function evaluations (NFE) can be chosen independently of output length, and early low-probability predictions can be revised. However, training or adapting models for diffusive generation has so far been computationally intensive and therefore expensive.

Our primary aim is therefore to devise a recipe optimized foremost for compute efficiency and reproducibility when converting AR models into DLMs, rather than to pretrain a new diffusion foundation model for every architecture. Efficiency matters for three connected reasons. First, full diffusion pretraining is financially and environmentally costly: under our standardized rental and electricity scenario, the reported LLaDA run corresponds to approximately \$675,000 and 44,100 kg CO₂e (Table 1). These are not the authors’ bill or measured emissions—operational carbon depends strongly on utilization, datacenter efficiency, location, and energy supply—but they illustrate the scale of the barrier (Strubell et al., 2019; Patterson et al., 2021). Second, a conversion that completes in hours on one widely available cloud instance makes hands-on DLM research accessible beyond laboratories with multi-node clusters. Third, applying the same inexpensive intervention to multiple existing model families makes it feasible to study how architecture and parent-model choice affect diffusion generation without repeating foundation-model pretraining for every comparison.

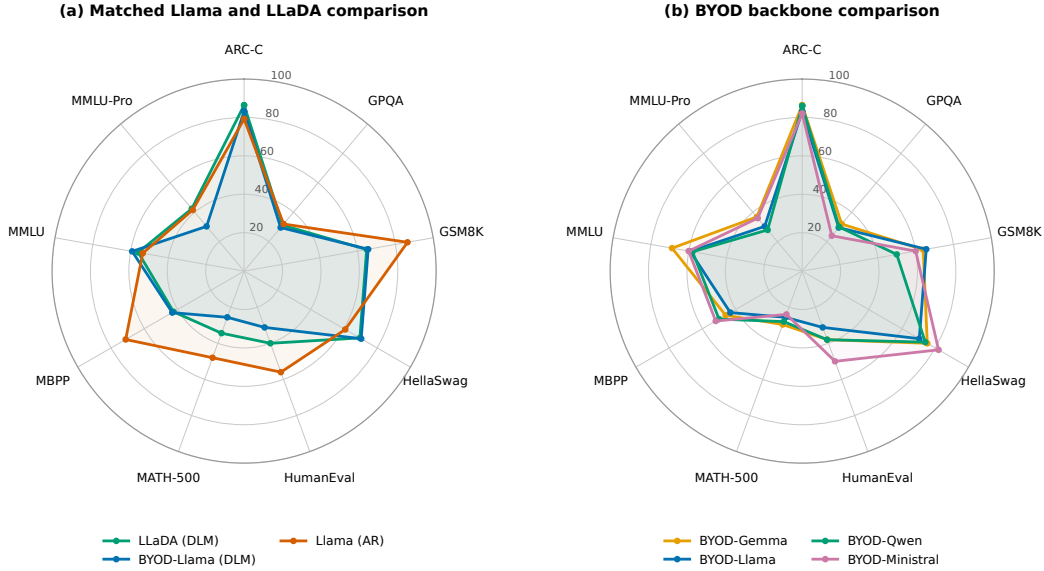


Figure 1: Task-level accuracy profiles from the matched evaluation. (a) LLaDA, BYOD-Llama, and the unchanged Llama AR parent. (b) The four primary BYOD conversions. Every spoke uses the same absolute 0–100% accuracy scale; benchmark-specific min–max scaling is deliberately avoided because it would make equal radial distances represent different accuracy changes. The polygon areas are descriptive visual summaries, not aggregate scores.

We study this question with BYOD, a generalizable conversion and evaluation pipeline for four instruction-tuned decoder families: Gemma 2 9B, Llama 3.1 8B, Qwen2.5 7B, and Ministral 8B. We selected the latest text-only instruction checkpoint from each family with a comparable parameter count. Multimodal variants are not used to avoid modality-specific training as an additional confounder. Every backbone receives the same bidirectional attention intervention, LoRA target modules, and masked-diffusion objective. This paired design separates conversion quality from the strength of the starting checkpoint. We directly compare our models against LLaDA 8B Instruct because its standard masked-diffusion objective, instruction-tuning setup, and overlapping benchmark suite make it the cleanest external control. We additionally report published DiffuLLaMA, Dream-Instruct, iLLaDA-Instruct, and Efficient-DLM values, but keep them protocol-separated: their additional attention, masking, scheduling, data, and inference techniques would confound a controlled comparison.

Our contributions are:

- a compute-efficient recipe that converts four instruction-tuned AR model families to masked diffusion using only a few single-GPU hours and LoRA parameter updates;
- a paired evaluation of each diffusion model and its frozen AR parent across general knowledge, reasoning, mathematics, and code benchmarks;
- a controlled analysis of continued training, denoising-step count, validation loss, perplexity and distinct n-grams; and
- publicly available training and inference code directly available through Google Colab and Github, and interactive demos for all four models, making DLM conversion and qualitative evaluation easily accessible.

2 RELATED WORK

Discrete diffusion models corrupt and reconstruct categorical sequences rather than adding Gaussian noise. D3PM formalized categorical corruption with transition matrices, including an absorbing process in which tokens transition to a dedicated mask state (Austin et al., 2021a). MDLM and

contemporaneous work showed that this masked formulation admits a simple noise-weighted cross-entropy objective, while RADD reparameterized the predictor in terms of conditional distributions over clean tokens (Sahoo et al., 2024; Shi et al., 2024; Ou et al., 2025). These approaches form the basis of masked diffusion language models (MDLMs). Uniform (or multinomial) diffusion, which instead corrupts tokens with samples from the ordinary vocabulary, is another option (Hooeboom et al., 2021); we use the simpler absorbing-mask denoising objective employed by LLaDA.

The authors of LLaDA were the first to show that masked diffusion can scale to an 8B-parameter general-purpose language model (Nie et al., 2025). Although its architecture follows Llama 3.1 8B, LLaDA was pre-trained from scratch using 2.3T tokens and 130,000 GPU-hours. AR-to-diffusion conversion instead amortizes knowledge already acquired by an AR checkpoint. DiffuLLaMA established this route through 65B tokens of full-model continued training at 7B scale (Gong et al., 2025). Dream adopts DiffuLLaMA’s shifted-prediction initialization, continues Qwen2.5 for 580B tokens, and adds context-adaptive token-level noise rescheduling (Ye et al., 2025). Efficient-DLM combines 500B tokens of continued training with blockwise attention and position-dependent masking (Fu et al., 2025), whereas iLLaDA returns to from-scratch diffusion pretraining at substantially larger scale (Nie et al., 2026). Diffusion conversion has also been demonstrated at larger sparse scales by RND1 (30B parameters), LLaDA2.0 (16B and 100B), and DiffusionGemma (25.2B, using uniform diffusion) (Chandrasegaran et al., 2025; Bie et al., 2025; DiffusionGemma Team, 2026).

Hybrid and blockwise models offer a different compromise between bidirectionality and deployability. Block Diffusion is autoregressive across blocks but diffusive within each block, enabling variable-length generation and inter-block KV caching (Arriola et al., 2025). Fast-dLLM v2 converts Qwen2.5 models using full-parameter fine-tuning with approximately 1B tokens and combines blockwise complementary masking with hierarchical caching (Wu et al., 2025). These systems retain more left-to-right structure than our fully bidirectional answer block, making them especially relevant to speed, caching, and arbitrary-length generation but not suited as direct controls for our intervention.

LAD first showed that LoRA can induce diffusion generation, using a hybrid objective that combines masking with token swaps, duplications, and span shifts (Kuiper et al., 2025; Hu et al., 2022). BYOD instead tests whether a simpler IID absorbing-mask objective can transfer reliably across model families. Other LoRA-based diffusion work addresses different goals: Uno learns diffusion-draft adapters while retaining the AR model as verifier, the dLLM framework uses LoRA to fine-tune already-diffusive checkpoints rather than for its Tiny-A2D conversion, and NaRA adapts existing diffusion models with noise-conditioned LoRA (Sahoo et al., 2026; Zhou et al., 2026; Wang et al., 2026b). Thus, their LoRA components augment already diffusively generating models, rather than converting AR model families into standalone diffusion models.

Table 1 places (estimates of) the training budgets of models of comparable size to our intended conversion targets side by side. For AR-to-diffusion conversion methods, these budgets exclude the substantial earlier cost of training the AR parent; for from-scratch methods they include diffusion pretraining. Dollar and operational-emissions columns are rental-equivalent estimates under one documented pricing and energy scenario, not the authors’ actual bills or measured footprints; Appendix F gives the complete calculation and its limitations.

Our study has explicitly chosen to hold the adaptation mechanism as constant as the architectures permit, repeat it across four families, and benchmark each converted model against the exact AR checkpoint from which it originated. This paired comparison is essential because it allows us to compare our results directly with the prior capability of the parent model.

3 METHOD

3.1 MASKED-DIFFUSION TRAINING OBJECTIVE

The conversion objective teaches the adapted model to recover clean answer tokens from randomly masked versions of the same response while conditioning on an uncorrupted instruction. Equation 1 formalizes this reconstruction loss and its correction for the sampled noise level; the following subsections describe the bidirectional attention intervention and the restricted set of parameters used to optimize it.

Table 1: Incremental training scale and standardized resource estimates for approximately comparable dense 3–9B diffusion systems. “Full” updates the main model weights; LoRA updates diffusion adapters. GPU-hours marked [†] are estimated from published model/data scale; unmarked values are reported. Costs are current cloud-rental equivalents and CO₂e is an operational scenario estimate (Appendix F), not a measured footprint. NR: insufficient information.

System	Diffusion-stage data	Updated weights	Hardware	GPU-h	Cost (USD)	CO ₂ e
BYOD (ours)	≤102.4M tokens	LoRA	1×RTX Pro 6000	3.2	\$11	0.9 kg
DiffuLLaMA 7B	65B tokens	Full	64×GH200	1,917 [†]	\$9,951	0.65 t
LLaDA 8B	2.3T tokens	Full	H800	130,000	\$674,830	44.1 t
Dream 7B	580B tokens + SFT	Full	NR	17,105 [†]	\$88,791	5.81 t
iLLaDA 8B	12T tokens + SFT	Full	NR	404,449 [†]	\$2,099,494	137.3 t
Efficient-DLM 8B	500B tokens	Full	128×H100	16,852 [†]	\$87,479	5.72 t
Uno 8B	7B tokens	LoRA	64×H200	3,840	\$19,933	1.30 t
Fast-dLLM v2 7B	~1B tokens	Full	64×A100	768	\$3,987	149 kg

Let p be an observed instruction prompt and $x_0 = (x_0^1, \dots, x_0^L)$ its clean response region, where L is the response-region length and x_0^i is its token at position i . For each example, we sample $t \sim \mathcal{U}[10^{-3}, 1]$ and independently replace each eligible response token with a single tokenizer-native mask representation with probability t . Denote the resulting sequence by x_t and the sampled mask positions by M_t . The model is trained with same-position cross-entropy,

$$\mathcal{L}(\theta) = -\mathbb{E}_{t, x_0, x_t} \left[\frac{1}{|E|} \sum_{i \in M_t} \log p_{\phi, \theta}(x_0^i | p, x_t) \right], \quad (1)$$

where E is the set of eligible answer-and-padding positions, $|E|$ is its cardinality, $M_t \subseteq E$ is the subset masked at noise level t , and $p_{\phi, \theta}$ is the model’s conditional token distribution. The expectation is over training examples, sampled noise levels, and their resulting corruptions; ϕ denotes the frozen parent parameters and θ the LoRA parameters. Prompt and chat-template tokens are always visible and never enter the loss. Training corruption is resampled online; validation corruption is deterministic from the example index and seed.

The clean terminating EOS is included as a genuine answer token. To teach the fixed-width generator to emit termination beyond the answer, dynamic batch padding is represented by repeated EOS tokens; these positions are also eligible for masking and loss. Thus padding is part of the learned denoising target rather than conventional ignored padding.

3.2 BIDIRECTIONAL CONVERSION AND TRAINABLE PARAMETERS

We replace the causal mask with an all-zero four-dimensional additive attention mask, so every prompt, answer, and EOS-padding position can attend to every position. The architecture and output vocabulary remain unchanged. This intervention makes right-hand response evidence available to masked positions; Section 5.2 tests whether the converted models use that access in practice. We attach LoRA adapters to every self-attention query and value projection, with rank $r = 1024$, scaling $\alpha = 1024$, and dropout zero. This high rank reflects the scale of the intervention: the adapters must learn a new conditional denoising process rather than a narrow downstream task. In LAD’s rank study, increasing rank from 512 to 1024 at 3B improved every reported benchmark and perplexity, whereas increasing from 1024 to 2048 at 8B gave no consistent further gain; our rank-selection experiments showed the same pattern. We therefore chose 1024 as a compromise between conversion capacity and training footprint (Kuiper et al., 2025). Backbone weights, token embeddings, the language-model head, biases, and normalization layers are frozen. Each model reuses an existing single-token string from its tokenizer (MASK for Gemma, Llama, and Qwen; <?> for Ministral), avoiding vocabulary expansion.

Table 2 reports the total and trainable parameter counts and measured conversion times. We hold LoRA rank fixed rather than matching the aggregate trainable count or fraction. Rank is the dimension of each low-rank update, so fixing it gives every targeted Q/V projection the same adapter bottleneck and preserves one architecture-agnostic recipe. Matching total counts would instead require architecture-specific, potentially nonuniform integer ranks because models differ in layer count, projection width, attention organization, and non-attention parameterization; matching a fraction

Table 2: Converted parent checkpoints, audited trainable parameter counts, and measured time for 25,000 updates. All trainable parameters are rank-1024 Q/V LoRA weights. Time is wall-clock hours on one RTX PRO 6000 GPU and therefore also equals aggregate GPU-hours.

Parent checkpoint	Total parameters	Trainable	Trainable (%)	Time (h)
Gemma 2 9B IT	10.732B	572.5M	5.33	4.0
Llama 3.1 8B Instruct	8.466B	436.2M	5.15	3.2
Qwen2.5 7B Instruct	7.939B	323.0M	4.07	3.1
Ministral 8B Instruct 2410	8.511B	490.7M	5.77	3.4

of total parameters would likewise not equalize adaptation capacity in the targeted modules. No convention makes these architectures fully apples-to-apples. Our choice avoids tuning the adapter budget separately for each backbone, but leaves 323.0–572.5M parameters (4.07–5.77%) trainable and may therefore influence the cross-family results. Equal-count and equal-fraction controls remain useful future ablations.

After training, we tested whether folding the LoRA layers into the parent projection would change the performance of the model, defined as: $W_{\text{merged}} = W + (\alpha/r)BA$, where W is the frozen base projection, A and B are the trained low-rank adapter matrices, r is their rank, and α is the LoRA scaling parameter. The resulting standalone checkpoint retains bidirectional inference but has the base architecture’s parameter count: Llama decreases from 8.466B adapter-loaded parameters to 8.030B merged parameters.

3.3 TRAINING DATA

All main models use the same training dataset and seed-selected ordering. Examples are retokenized with each parent model’s native chat template and tokenizer. Only the current answer region is corrupted. The dataset builder groups sources into general instruction following, multiple-choice reasoning, mathematics, and code, deduplicates normalized prompts, and blocks exact normalized matches with represented benchmark validation/test prompts.

The four primary models see the same dataset prefix. The continued Llama condition initializes from the primary adapter and uses the next non-overlapping segment of the stable shuffle. It resamples corruption masks and restarts AdamW and its learning-rate schedule; it is therefore continued adapter training, not a restoration of optimizer state. Exact stage sizes are consolidated in Appendix B.

3.4 OPTIMIZATION

We use AdamW with peak learning rate 10^{-5} and train with BF16 master weights plus Transformer Engine HYBRID FP8 execution. All reported conversions ran on a single NVIDIA RTX PRO 6000 Blackwell Server Edition in Google Colab, without multi-GPU or multi-node parallelism; the code, memory strategy, and adapter scope were explicitly designed around this constraint. Appendix B gives the complete optimization, checkpointing, validation, and reproducibility configuration.

3.5 DIFFUSION GENERATION

Generation begins from a fully masked answer block. At each reverse step the model predicts every masked position, after which the least-confident positions are remasked for reconsideration; newly accepted positions become permanent unmasked.

4 EVALUATION

4.1 PAIRED MODELS AND BASELINES

For BYOD-Gemma, -Llama, -Qwen, and -Ministral we compare the diffusion version with greedy generation from its unchanged AR parent. We additionally compare the BYOD-Llama-50k longer-trained variant. LLaDA 8B Instruct is evaluated in the same evaluation harness as an external diffusion baseline. Additionally, we compare our results with values transcribed from the LLaDA

paper/evaluation suite. The latter are contextual references rather than directly matched measurements. The primary factual and code benchmarks use greedy token selection and low-confidence remasking. Open-ended evaluation instead samples at temperature 0.7 and delays acceptance of EOS/EOT predictions. Complete task-specific diffusion and AR settings are reported in Appendix C.

4.2 BENCHMARKS

Our benchmark set contains the validation/test sets of MMLU, MMLU-Pro, HellaSwag, ARC-Challenge, GSM8K, MATH-500, GPQA-Main, HumanEval, and MBPP (Hendrycks et al., 2021a; Wang et al., 2024; Zellers et al., 2019; Clark et al., 2018; Cobbe et al., 2021; Hendrycks et al., 2021b; Rein et al., 2023; Chen et al., 2021; Austin et al., 2021b). We score multiple-choice tasks by option label, GSM8K by its final number, MATH by its normalized final boxed expression, and code tasks by whether the generated program passes the supplied tests in an isolated subprocess.

4.3 FLUENCY

We use perplexity and distinct 1/2/3-gram percentages as a proxy for fluency. We calculate these based on the text generated by each diffusion model and its AR parent as answers on 100 open-ended questions. Perplexity is measured using the larger Microsoft Phi-4 14B model, which scores each generated answer. Our primary aggregate is token-weighted perplexity, $\exp(\sum_j \text{NLL}_j / \sum_j N_j)$, so each scored token contributes equally. As perplexity does not penalize loss of sample diversity (Turok et al., 2026), we also compute sliding model-token Distinct-1/2/3 after excluding special tokens and report them as percentages. Higher Distinct- n means a larger fraction of unique n -grams, and thus a more diverse text.

4.4 BIDIRECTIONAL ATTENTION

We test whether the converted diffusion actually allow for bidirectional information flow with a simple right-context copy task (Appendix E). In this task, most of the response of the model is already pre-filled. Its task is to fill in the last missing word by copying the word shown later in the response. This would not be possible without bidirectional attention, as a causal model cannot change its prediction at an earlier position based on a token later in the sequence.

Right-context copy example

System: *empty*

User: Fill the earlier missing word by copying the word shown later in the response.

Partially observed assistant response:

Required prediction at [MASK]: apple

Reconstructed assistant response:

Earlier word: apple. Later word to copy: apple.

Here [MASK] denotes the model’s tokenizer-native mask token. The model predicts only the masked position; the reconstructed response is shown to make the expected result explicit.

5 RESULTS

5.1 BENCHMARK ACCURACY AND BASE-MODEL DEPENDENCE

Table 3 reports the matched-harness task results and their unweighted macro average.

Gemma is the strongest converted backbone, with a 53.0% macro average, followed by Ministral at 52.2%; both exceed LLaDA’s 51.9% in the same harness. The matched AR parents nevertheless outperform their converted counterparts on average, although individual tasks differ. Gemma is also the largest parent and receives the largest absolute LoRA update budget, offering two plausible advantages in this comparison. Its AR parent is strong, but Qwen’s higher AR macro average shows that parent accuracy alone does not determine conversion quality. Notably, the earlier two-stage Llama run does not uniformly improve downstream accuracy, motivating checkpoint-wise evaluation rather than selecting solely by denoising loss. Section 5.3 separately examines fluency across an

Table 3: Exact-match/pass@1 accuracy in our matched harness. D: converted diffusion model; AR: unchanged parent. Llama-50k shares the Llama-25k AR parent. Among common-harness diffusion columns, bold marks the best result and underline the best converted BYOD result (ties retained); published LLaDA is protocol-separated and excluded from highlighting.

Task	Gemma 25k		Llama		Qwen 25k		Ministral 25k		LLaDA	
	D	AR	D-25k	D-50k	AR	D	AR	D	AR	Ours Published
<i>General knowledge and reasoning (multiple choice)</i>										
ARC-C	.864	.888	.832	.792	.792	.860	.920	.820	.740	.864 .885
GPQA	.320	.280	.296	.272	.320	.300	.400	.240	.340	.312 .333
HellaSwag	.752	.696	.704	.688	.608	.740	.780	.820	.700	.696 .746
MMLU	.688	.648	.592	.608	.536	.580	.660	.600	.520	.560 .655
MMLU-Pro	.368	.448	.304	.304	.416	.280	.400	.360	.320	.424 .370
<i>Math (open ended)</i>										
GSM8K	.640	.872	.656	.560	.864	.500	.900	.600	.860	.648 .694
MATH-500	.296	.600	.256	.216	.480	.280	.780	.240	.520	.344 .319
<i>Code generation (open ended)</i>										
HumanEval	.380	.740	.312	.336	.560	.380	.960	.500	.940	.400 .494
MBPP	.460	.740	.432	.512	.712	.500	.820	.520	.720	.424 .410
Average	.530	.662	.487	.476	.588	.491	.736	.522	.629	.519 .545

uninterrupted 50k-update run. Published values from other diffusion systems are provided separately in Appendix A because their protocols are not matched to ours.

Figure 1 makes the task profiles visible without collapsing them into a macro average. Panel (a) contrasts BYOD-Llama with both its unchanged AR parent and LLaDA. The AR parent remains substantially stronger on generation-heavy mathematics and code tasks, whereas the two DLMs have broadly similar profiles. Panel (b) shows that the relative strengths of the four BYOD backbones are task dependent.

LLaDA scores 51.9% on the nine-task unweighted average in our harness, compared with its published average of 54.5%. This is not an exact reproduction because prompts, subsets, generation settings, answer extraction, and code execution may differ. We therefore use the common-harness comparison as primary and report published results separately in Appendix A. LLaDA is our only directly tested external DLM because its response-masking objective and benchmark suite provide the cleanest comparison; Dream, iLLaDA, and Efficient-DLM introduce additional differences in data scale, masking, attention, and inference procedures.

5.2 RIGHT-CONTEXT COPY TEST

The right-context copy test asks whether a model can copy a later visible word into an earlier masked position (Appendix E). With the clue present, exact match is 95% for Gemma, 75% for each Llama checkpoint, 15% for Qwen, 30% for Ministral, and 90% for LLaDA; every model scores 0% after the clue is removed. Exact copying therefore varies substantially. However, the clue changes the greedy prediction in 100% of Gemma, long-run Llama, Ministral, and LLaDA examples and 95% for Llama-25k, showing strong right-context use even where instruction following is imperfect. Qwen changes its prediction in only 15% of examples and is the clear exception.

5.3 FLUENCY WITH REDUCED DENOISING STEPS

Table 4: Open-ended generation at a 128-token output length and 64 NFE for diffusion models. PPL is token-weighted Phi-4 perplexity (lower is better); D-1/2/3 are mean per-answer sliding Distinct- n percentages over model-tokenizer tokens (higher is less repetitive). AR parents generate the same output length autoregressively.

Model family/checkpoint	Diffusion				AR parent			
	PPL↓	D-1%↑	D-2%↑	D-3%↑	PPL↓	D-1%↑	D-2%↑	D-3%↑
Gemma 2 9B, 25k	8.27	45.5	75.2	85.8	3.01	63.4	89.3	95.8
Llama 3.1 8B, 25k	10.50	52.7	80.2	90.5	2.49	61.0	89.0	94.8
Llama 3.1 8B, 50k	9.53	49.6	79.1	89.2	2.49	61.0	89.0	94.8
Qwen2.5 7B, 25k	11.12	50.9	80.4	90.4	2.23	69.5	94.0	98.6
Ministral 8B, 25k	9.42	49.3	78.6	88.8	2.31	65.1	91.3	97.1
LLaDA 8B Instruct	4.07	51.4	82.3	91.6	—	—	—	—

Increasing the denoising budget strongly improves this diagnostic. In the latest open-ended sweep, Llama-25k changes from 19.381 at 32 NFE to 10.503 at 64 and 5.081 at 128; Gemma-25k changes from 14.560 to 8.273 and 4.334. We call settings with fewer denoising network evaluations than generated tokens ($\text{NFE} < L$) the *parallel decoding regime*; unlike AR decoding, they permit more than one token to be finalized per model evaluation. Perplexity values are meaningful only under matched output length, sampler, temperature, and denoising budget.

We also benchmarked the saved Llama adapter before and after merging (Appendix G). Merging removes 436.2M deployed parameters while changing token-weighted perplexity from 19.381 to 19.283 (-0.51%) at 32 NFE and from 3.753 to 3.720 at 128 NFE; Distinct-1/2/3 at 32 NFE change from 49.5/78.8/89.9% to 49.8/80.7/91.8%. On the nine-task subsets, the unweighted accuracy mean changes from 51.3% to 49.3%, with mixed task-level differences. Across these metric families, merging preserves the model’s overall behavior without a systematic open-generation penalty.

5.4 FURTHER TRAINING

Figure 2 shows checkpoint-wise generation diagnostics. Improvements are not monotonic at every budget. In the uninterrupted 50k-update run, 32-NFE perplexity reaches its minimum of 16.593 at 40k and ends at 17.002, while Distinct-1 ends at 43.5%. The 64-NFE sweep reaches its minimum of 9.089 at 35k and ends at 9.534 with 49.6% Distinct-1; at 128 NFE, the corresponding values are 4.802 at 35k and 4.830 with 50.7% Distinct-1 at 50k. Perplexity fluctuations are largest at low NFE despite the continuously decreasing validation loss in panel (c). Checkpoints 30k and 45k terminate earlier under all three decoding budgets: their mean scored lengths are 75.3/74.5 tokens at 32 NFE, 77.6/78.4 at 64 NFE, and 85.7/81.5 at 128 NFE, whereas their respective neighboring checkpoints span 95.3–102.4, 101.1–102.3, and 104.9–110.7 tokens. A targeted 32-NFE replication with a second sampling seed produces the same behavior: 30k/45k obtain 21.417/20.678 perplexity and mean lengths of 72.6/73.8 tokens, close to 22.934/22.202 and 75.3/74.5 with the original seed. Recurrence across denoising budgets and sampling seeds therefore strongly supports checkpoint-specific termination behavior rather than a single anomalous generation. Under the same generation protocol, the unchanged AR parent obtains 2.639 perplexity and 62.1% Distinct-1. The factual-task macro average of the earlier two-stage model changes only from 48.7% at 25k to 47.6% at 50k. Further uninterrupted training may therefore affect low-budget parallel decoding more strongly than factual capability, but does not yield monotonic gains.

6 DISCUSSION

The central empirical result is that useful AR-to-diffusion conversion can be made an hours-scale, single-GPU procedure. High-rank attention-only LoRA is sufficient to turn four independently developed causal instruction models into generators that reconstruct answers from a fully masked block. Together with the right-context copy test and the retained benchmark performance—on par with LLaDA in our common-harness macro average—this indicates that the models acquire functional bidirectional generation while preserving substantial knowledge from their AR parents. A capable paid cloud GPU is still required, so this represents broader rather than zero-cost access.

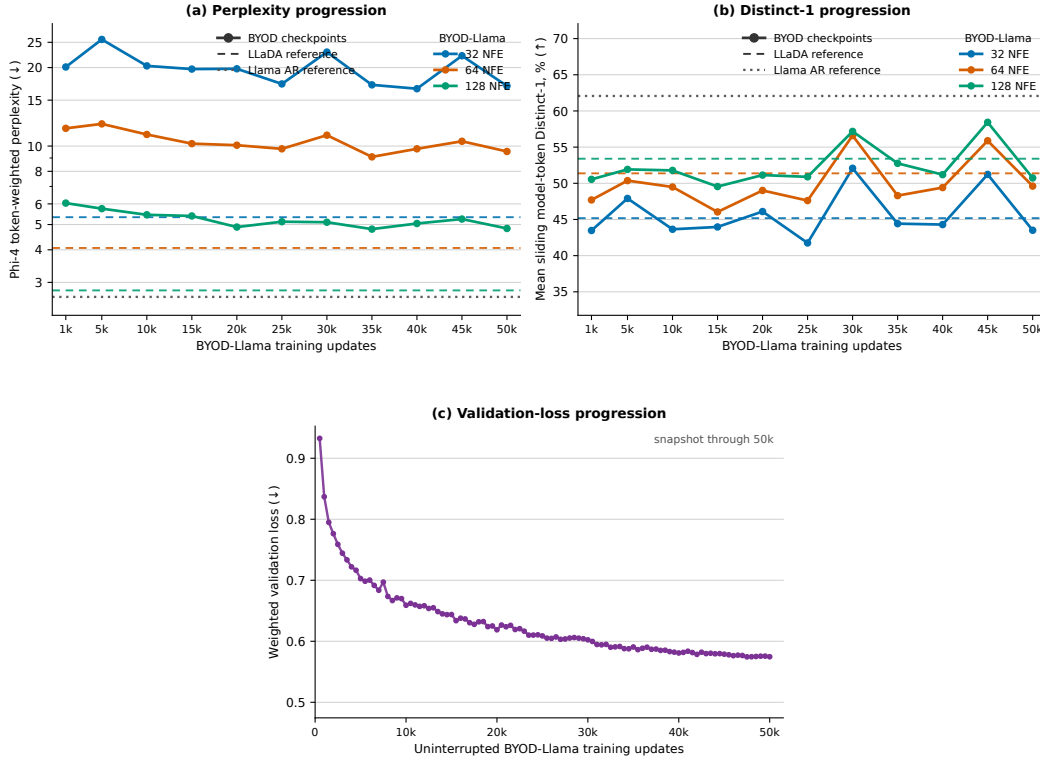


Figure 2: BYOD-Llama training diagnostics. (a) Token-weighted Phi-4 perplexity for 128-token generations, shown on a logarithmic y-axis. (b) Mean sliding model-token Distinct-1 for the same generations, reported as a percentage. Each NFE series covers all eleven checkpoints from the uninterrupted llama-3.1-8b-mask-long run. Color-matched dashed lines are fixed LLaDA references; the dotted gray line is the unchanged Llama AR parent. (c) Weighted validation loss from the uninterrupted run through 50k updates. Lower is better in (a,c), and higher is better in (b).

The backbone comparison also shows that conversion quality is not architecture independent. Gemma produces the strongest task average among the converted models. Its larger parent model and larger absolute LoRA parameter budget plausibly provide more capacity for learning the new denoising behavior, while its capable AR starting point supplies knowledge worth retaining. These factors cannot be disentangled here: Qwen has the strongest AR average but does not yield the strongest conversion, so parent capability is important but insufficient on its own.

Accessibility here concerns evaluation as well as training. Our streamlined Google Colab workflow installs the required software, offers a choice of supported AR parents, and provides live training feedback, allowing the conversion to be reproduced or modified without a local accelerator cluster or bespoke training infrastructure. We also provide full-precision interactive demos for every converted model. These are part of the scientific evaluation rather than merely presentation artifacts: perplexity and Distinct- n expose only narrow aspects of generation, whereas grammaticality, coherence, characteristic failures, and the denoising process are often easiest to assess through direct interaction. Releasing weights and source code without a working inference path leaves readers with substantial setup work and makes it less likely that they will qualitatively inspect each new model. The demos remove much of that friction and make such scrutiny an accessible part of the reproducibility package.

Our public demos implement EOS-aware early stopping—halting when the pre-EOS answer is identical in two consecutive iterations, together with delayed EOS/EOT retention—which allows actual inference time to fall below the configured fixed-canvas budget; related EOS-led variable-length or early-stopping mechanisms are used by dLLM-Var, VoidPadding, and audio-conditioned diffusion ASR, while JoT instead detects convergence per token (Yang et al., 2025; Liu et al., 2026; Wang et al., 2026a; Kohut et al., 2026).

The merge result also shows that LoRA’s parameter-count overhead is not permanent: the deployed Llama returns from 8.466B to 8.030B parameters without a material open-generation degradation in this small matched comparison (Appendix G). Algebraically folding $(\alpha/r)BA$ into W removes both the stored adapter branch and its extra low-rank inference matrix multiplications. Consequently, using hundreds of millions of high-rank LoRA parameters during conversion need not make the final generator larger or more expensive per network evaluation than the base architecture. This is not model compression—the full base-sized checkpoint remains—and diffusion decoding can still require many network evaluations; more prompts, deterministic parity checks, and backbones are needed to bound small numerical or sampling differences after merging.

Conversion is not free, and successful diffusion behavior is not yet equivalent to the fluency of a purpose-trained DLM. First, 323–573M trainable parameters are parameter-efficient relative to full-model training but substantially larger than conventional low-rank task adapters. Second, although knowledge retention is competitive, converted-model perplexity and Distinct-1/2/3 generally remain worse than both the AR parents and LLaDA. The gap is largest in low-NFE parallel decoding, precisely where diffusion offers its most interesting efficiency advantage. Longer uninterrupted training may narrow it, but the current progression is not monotonically improving and a lower learning rate alone did not resolve the issue. The synchronized shortening at 30k and 45k across three denoising budgets and two sampling seeds indicates that termination calibration can drift during training even while validation loss improves; the accompanying increase in Distinct-1 should therefore not be interpreted as an unqualified quality gain. Closing the gap may instead require a larger effective batch, a revised denoising or loss target, or architectural adaptation beyond query/value projections—for example, testing key/output projections, MLP adapters, or selectively trainable normalization layers. These are hypotheses for controlled ablations, not conclusions from the present runs. Finally, EOS-padding supervision makes the fixed-width objective different from a pure answer-only likelihood and deserves explicit ablation.

Perplexity alone is insufficient for selecting a diffusion generator. DiffuLLaMA pairs external-model perplexity with distinct bigrams to check that low perplexity was not obtained through repetition (Gong et al., 2025). More fundamentally, external generative perplexity reflects a separate AR model’s preferences and ignores diversity, so repeated high-probability text can score deceptively well (Turok et al., 2026). Theory likewise distinguishes token-level perplexity from whole-sequence correctness: a masked diffusion model may approach favorable perplexity with few steps while still requiring a step count that scales with sequence length for low sequence error (Feng et al., 2025). We therefore triangulate external perplexity with Distinct-1/2/3, task accuracy, and generated samples. Distinct- n is itself only a repetition diagnostic, so grammaticality, coherence, and actual usability ultimately require systematic qualitative inspection rather than replacement by any single automatic score.

The literature comparison sharpens the trade-off. DiffuLLaMA and Dream reuse AR initialization but update the full model for 65B and 580B tokens, respectively; Efficient-DLM invests 500B tokens and architectural training choices specifically designed to retain causal-model behavior and enable cached block decoding (Gong et al., 2025; Ye et al., 2025; Fu et al., 2025). iLLaDA’s strong reported results come from 12T-token diffusion pretraining plus repeated large-scale SFT, a fundamentally different compute regime (Nie et al., 2026). Fast-dLLM v2 and SDLM show that retaining blockwise or sequential structure can reduce conversion cost while enabling practical caching and variable-length generation (Wu et al., 2025; Liu et al., 2025); our complementary question is how far a fully bidirectional model can be converted with adapter-only updates. Uno is closer to our parameter-efficient philosophy because it also trains LoRA diffusion weights, but its goal is different: an AR verifier guarantees samples from the original AR distribution, whereas BYOD seeks an independently usable fully bidirectional generator (Sahoo et al., 2026). Thus the higher published scores in Appendix A do not isolate a single superior masking objective; they bundle substantially different data budgets and modeling or decoding machinery.

The incremental scale is central to the contribution. One primary BYOD conversion processes orders of magnitude fewer token positions than the full-model conversion systems in Table 1, while leaving most parent parameters frozen. This comparison does not erase the sunk cost of AR pretraining, nor prove a proportional reduction in energy or emissions. It does show that an existing foundation checkpoint can be repurposed in a few hours without repeating pretraining or reserving a large cluster. Economically, this turns architecture comparison from multiple foundation-model training projects into repeatable adapter experiments. Environmentally, it avoids duplicating much of the operational

compute, although direct power measurements and the time- and location-specific electricity mix remain necessary for a defensible carbon comparison (Strubell et al., 2019; Patterson et al., 2021).

This is also why we do not claim that Appendix A is a leaderboard. Our direct LLaDA run holds prompts, subsets, extraction, and generation code fixed. The remaining columns are literature context, useful for locating the attainable performance range but unsuitable for causal comparison. A future study can add these systems to the shared harness one at a time, provided their task-specific inference procedures are faithfully reproduced—a requirement emphasized by the dLLM framework (Zhou et al., 2026).

7 LIMITATIONS

Only one training seed is available per model, so architecture and random-seed effects are not separated. The four backbones differ in tokenizer, pretraining data, alignment, attention implementation, and nominal parameter count. We fixed LoRA rank rather than total trainable parameters or trainable fraction; consequently, Gemma is both the largest parent and the model with the largest absolute adapter budget. Its leading benchmark result therefore cannot be attributed cleanly to architecture, parent quality, model scale, or conversion efficiency. Although chosen as the latest comparable text-only releases when the study was designed, the backbones are not guaranteed to remain the latest checkpoints at publication time. Phi-4 perplexity favors text that its own AR distribution considers probable, ignores diversity, and is not a diffusion likelihood. Distinct- n detects local repetition but not factuality, grammar, or global coherence. Published LLaDA numbers are not directly comparable to our harness. We do not report measured wall-clock inference speedups: $\text{NFE} < L$ reduces sequential model calls but does not automatically imply lower latency because each call processes the full answer block. Finally, we did not instrument the runs for energy consumption or carbon intensity. Single-GPU execution and lower incremental training volume support accessibility and compute-efficiency claims, not a quantified sustainability claim.

8 CONCLUSION

BYOD provides a common, efficient route from instruction-tuned AR checkpoints to masked diffusion generators across Gemma, Llama, Qwen, and Ministral. The experiments establish that diffusive behavior, functional bidirectional conditioning, and substantial parent-model knowledge can be obtained in a few hours on one Google Colab GPU. They also expose the remaining challenge: fluency and diversity lag behind AR parents and LLaDA, especially when NFE is below output length. The next step is therefore not merely more compute, but determining whether longer uninterrupted training, larger effective batches, alternative objectives, or a broader adapter placement can close this gap without sacrificing BYOD’s accessibility. In its present form, BYOD makes that experimentation feasible for researchers who cannot retrain a foundation model or provision a large accelerator cluster.

REPRODUCIBILITY STATEMENT

Training and evaluation code is available at <https://github.com/RuurdKuiper/lad-generics>. Public model cards and interactive demos for all four conversions are available under <https://huggingface.co/Ruurd>; each model card links to its corresponding full-precision ZeroGPU demo. The main text specifies the objective, trainable modules, data volume, optimization, generation, benchmarks, and scoring. The accompanying artifacts include resolved configurations, parameter audits, checkpoint identifiers, loss logs, generated samples, and benchmark manifests.

ETHICS STATEMENT

This study uses existing language models and public benchmark or instruction datasets and does not involve human subjects. Converted models inherit factual errors, biases, unsafe behaviors, licensing constraints, and memorization risks from their parent checkpoints and training data. We report aggregate task performance rather than claiming safety or deployment readiness; release artifacts should preserve upstream licenses and document the data provenance and intended research use.

AI USE STATEMENT

Generative AI tools were used to assist software development and debugging, design and review of experiments, creation of analysis scripts, interpretation of results, literature discovery and summarization, and drafting and editing this manuscript. The authors manually reviewed AI-assisted code and prose, checked numerical claims against saved experimental artifacts, and verified literature claims and citations against primary sources. No benchmark value was entered solely from an unsupported model response. The authors take responsibility for the final content, claims, code, and artifacts.

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A PROTOCOL-SEPARATED LITERATURE RESULTS

Table 5 places our primary Llama conversion beside values reported by other diffusion-model papers (Nie et al., 2025; Ye et al., 2025; Nie et al., 2026; Fu et al., 2025). The BYOD column uses our common harness, whereas every other column uses its source paper’s evaluation protocol. Differences in prompts, subsets, generation budgets, answer extraction, and execution environments prevent direct ranking; the table is included only to provide scale and task coverage. NR means that the source did not report a value for that task.

Table 5: Our Llama result and protocol-separated published diffusion-model accuracies. Bold marks the largest reported value in each row; it does not imply a controlled comparison. No cross-task averages are shown because task coverage differs by model.

Task	BYOD-Llama (ours)	LLaDA	Dream-Inst.	iLLaDA-Inst.	Efficient-DLM 8B
ARC-C	.832	.885	NR	NR	.619
GPQA	.296	.333	.330	NR	NR
GSM8K	.656	.694	.810	.890	.883
HellaSwag	.704	.746	NR	NR	.725
HumanEval	.312	.494	.555	.659	.683
MATH	.256	.319	.392	.567	.501
MBPP	.432	.410	.588	.580	.610
MMLU	.592	.655	.670	.716	.772
MMLU-Pro	.304	.370	.433	.523	NR

B COMPLETE TRAINING CONFIGURATION

Table 6 records the common resolved configuration of the four primary conversions. The continued Llama condition repeats the same stage from the first adapter checkpoint on the next non-overlapping data segment, with newly sampled masks and a freshly initialized optimizer and schedule; its checkpoints are spaced every 5,000 updates.

C COMPLETE INFERENCE CONFIGURATION

Table 7 separates task-specific diffusion budgets from the training configuration and gives the evaluation sample count in one place. Unless overridden in the table, diffusion evaluation uses the official LLaDA-style fixed-transfer sampler, greedy predictions (temperature 0), low-confidence remasking, classifier-free guidance scale 0, one full-answer block, and no repetition penalty. Delayed EOS/EOT means that a predicted terminator is assigned the lowest retention confidence and may therefore be predicted but is accepted late; HumanEval instead suppresses terminator logits. The

Table 6: Full common training configuration. Model-specific parent checkpoints, mask strings, and parameter counts appear in Table 2.

Setting	Value	Setting	Value
Examples per stage	400,000	Optimizer updates per stage	25,000
Maximum sequence length	256	Dynamic padding multiple	16
Batch size	16/16	Gradient accumulation	1
Optimizer	AdamW	Peak learning rate	10^{-5}
Schedule	cosine	Warmup	100 updates
Weight decay	0.01	Maximum gradient norm	0.5
Training precision	FP8 HYBRID	Master/evaluation precision	BF16
Gradient checkpointing	disabled	Selected-logit optimization	enabled
Corruption	IID mask-only	Noise time	$t \sim \mathcal{U}[0.001, 1]$
Mask-eligible region	answer + EOS padding	Loss region	corrupted answer + EOS padding
Answer/padding loss weights	0.9/0.1	Genuine answer EOS	included
LoRA targets	query/value projections	LoRA rank/scaling/dropout	1024/1024/0
Trainable normalization	disabled	Quantization	none
Validation	200 examples/250 updates	Checkpoint interval	1,000 updates
Data-loader workers	4	Prefetch factor	4

Table 7: Task-specific masked-diffusion inference settings in the validation notebook. NFE is the number of denoising network evaluations.

Task	Output tokens	NFE	Block length	Temperature	Terminator handling	Samples
Open-ended	128	128/64/32	128	0.7	delayed retention	125
MMLU	64	64	64	0	ordinary	125
MMLU-Pro	128	128	128	0	delayed retention	125
HellaSwag	64	64	64	0	ordinary	125
ARC-Challenge	64	64	64	0	ordinary	125
GSM8K	256	256	256	0	delayed retention	125
MATH-500	256	256	256	0	delayed retention	125
GPQA-Main	128	128	128	0	delayed retention	125
HumanEval	256	256	256	0	delayed retention	125
MBPP	128	128	128	0	delayed retention	125
Right-context copy	1 masked word	1	–	0	–	20

open-ended diagnostic uses temperature 0.7. Archived run manifests take precedence over mutable notebook defaults.

Autoregressive parents are decoded greedily. Their maximum-new-token budgets are reported separately because an AR token budget is not an NFE budget and should not be read as computational equivalence.

Table 8: Maximum new tokens for paired greedy AR evaluation.

Task	Tokens	Task	Tokens
Open-ended	128	MMLU	64
MMLU-Pro	2,048	HellaSwag	64
ARC-Challenge	64	GSM8K	256
MATH-500	256	GPQA-Main	128
HumanEval	256	MBPP	128

D INTERACTIVE DENOISING TRAJECTORIES

Figures 3 and 4 illustrate how the public BYOD-Gemma demo exposes the parallel refinement process under random and confidence-guided remasking. All panels use the displayed Amsterdam prompt, the helpful-assistant system prompt, temperature 0.7, top- k 3, initial noise fraction 1.0, delayed EOS/EOT retention, permanent locked retention, and no visualization delay. Early stopping is disabled and block length equals the new-token canvas, ensuring that every panel displays its complete denoising budget. The demo randomizes the hidden sampling seed for every request; we resample the 64-NFE conditions to show readable positive examples, so these trajectories illustrate behavior rather than constitute a controlled quality comparison.

Denoising step 64/64 · 46 output tokens

Amsterdam is the capital of the Netherlands and one of the most popular cities in the world. It is known for its canals, museums, and vibrant nightlife scene. It is also known for its liberal culture and bike-friendly environment.

(a) 64 NFE and 64 tokens.

Denoising step 32/32 · 42 output tokens

Amsterdam is the capital of the country of Netherlands. It is a for-city and is known for the canals, bicycles, and art museums. It is also one of provinces of the Netherlands of the Netherlands.

(b) 32 NFE and 64 tokens.

Denoising step 16/16 · 36 output tokens

Amsterdam is the capital and largest of the Netherlands. It is known for its canals, museums, and and nightlife.. It is a major and center of art and culture..

(c) 16 NFE and 64 tokens.

Figure 3: Confidence-guided BYOD-Gemma-2-9B trajectories at decreasing inference budgets on a 64-token canvas. Red-to-green coloring records each retained token’s sampling probability.

Denoising step 64/64 · 64 output tokens

Amsterdam is the largest and capital city of the Netherlands and is the most populated. It is known for its vibrant culture, rich history, bicycle-friendly culture, and its number of museums and attractions, such as the Rijks, the houses and canals, and the Anne Frank House. It is also a major center for tourism

(a) Random remasking; prediction-iteration coloring.

Denoising step 64/64 · 46 output tokens

Amsterdam is the capital of the Netherlands and one of the most popular cities in the world. It is known for its canals, museums, and vibrant nightlife scene. It is also known for its liberal culture and bike-friendly environment.

(b) Confidence-guided remasking; prediction-iteration coloring.

Figure 4: Remasking comparison for BYOD-Gemma-2-9B at 64 NFE on a 64-token canvas. Blue intensity records the iteration at which each retained token was predicted.

E RIGHT-CONTEXT COPY TEST

This deterministic test uses ordinary target words. Each response has the form “Earlier word: [MASK]. Later word to copy: apple.”, with a different target word in each example. The target is absent from the instruction and left context and appears explicitly after the mask. All model tokens belonging to the target word are masked simultaneously, and the converted model predicts them in one forward pass; no iterative sampling or generation settings are involved.

We score the same masked position twice: once with the later copy clue and once after deleting that clue. We report exact match for both conditions, their difference, and whether adding the clue changes the greedy prediction at all. Under causal attention, changing or deleting tokens to the right cannot affect logits at an earlier position, so both the exact-match gain and prediction-change rate must be zero. The latter separates use of right context from the stricter ability to follow the exact-copy instruction. This is a functional attention check, not a benchmark of reasoning or general infilling quality.

Table 9: Right-context copy results. “With clue” predicts the earlier mask while the word is visible later; “without clue” removes that later word. Gain is the exact-match difference.

Model	With clue EM	Without clue EM	EM gain	Prediction changed
Gemma 2 9B, 25k	95%	0%	+95 pp	100%
Llama 3.1 8B, 25k	75%	0%	+75 pp	95%
Llama 3.1 8B, long-run best	75%	0%	+75 pp	100%
Qwen2.5 7B, 25k	15%	0%	+15 pp	15%
Minstral 8B, 25k	30%	0%	+30 pp	100%
LLaDA 8B Instruct	90%	0%	+90 pp	100%

F COST AND OPERATIONAL-EMISSIONS ESTIMATION

Table 1 compares the cost of training a model that produces diffusion behavior. It should thus be taken into account that some of these models build on an AR parent, whereas others are trained from scratch. It also excludes exploratory runs, failed jobs, dataset construction, evaluation, storage, and inference. The estimates must therefore not be interpreted as total project or lifecycle footprints.

For BYOD we use the observed 3.2 GPU-hours for the conversion of a Llama 3.1 8B model for diffusive generation. For a prior work that reports GPU-hours, we retain that value. Where a paper reports model size but not time, we approximate training compute as $C \approx 6ND$ FLOPs (Hoffmann et al., 2022), where C is total training compute, N is the number of model parameters, and D is the number of processed training-token positions. We then calculate

$$G_{\text{est}} = \frac{6ND}{0.40 (989 \times 10^{12}) 3600}, \quad (2)$$

where G_{est} is aggregate estimated GPU-hours, 0.40 is the assumed model FLOPs utilization (MFU), 989×10^{12} is the NVIDIA H100’s dense BF16 Tensor Core throughput in FLOP/s (NVIDIA, 2023), and 3600 converts seconds to hours. The 40% MFU assumption is consistent with the 39.0–45.0% BF16 MFUs reported by FP8-LM for GPT-style 7B, 13B, and 175B training on H100 GPUs (Peng et al., 2023). Actual utilization varies considerably with architecture, sequence length, parallelization, communication, activation recomputation, and implementation efficiency. The assumption is therefore a standardized order-of-magnitude convention rather than a reconstruction of any authors’ cluster. It yields the daggered values in Table 1. The Dream and iLLaDA estimates cover their main 580B- and 12T-token stages but omit the smaller instruction-tuning additions, so they are lower bounds.

Table 10 calibrates the same calculation against studies that report both a usable token count and elapsed accelerator-hours (Nie et al., 2025; Wu et al., 2025). For LLaDA, it predicts approximately 77,519 GPU-hours, compared with 130,000 reported H800 GPU-hours, corresponding to an implied utilization of 23.9%. Thus 40% can be optimistic about achieved utilization and correspondingly conservative with respect to estimated accelerator-hours, rental cost, and emissions. Fast-dLLM v2 displays a still larger gap. Its specialized training concatenates clean and noised views and

uses a nonstandard attention pattern, illustrating that an “implied MFU” from $6ND$ is a calibration diagnostic, not a measured hardware-utilization statistic.

Table 10: Calibration of the $6ND$ estimator against reported accelerator-hours. Predictions use 40% MFU and the stated accelerator’s dense-BF16 peak (989 TFLOP/s H100-class proxy for H800; 312 TFLOP/s for A100) (NVIDIA, 2023; 2021). “Implied MFU” is the utilization that would reconcile $6ND$ with reported time; it is not a measured MFU.

System	Parameters	Tokens	Pred. GPU-h	Reported GPU-h	Implied MFU
LLaDA	8B	2.30T	77,519	130,000	23.9%
Fast-dLLM v2	7B	1.31B	123	768	6.4%

We convert BYOD time to a rental-equivalent cost using the September 2026 AWS US-East on-demand price of \$3.36 per hour for a one-GPU `g7e.2xlarge`, which contains the same 96-GB NVIDIA RTX PRO Server 6000 accelerator used in our Colab run (Amazon Web Services, 2026b). For other systems we use one consistent current major-cloud proxy: \$5.191 per accelerator-hour for an AWS `p5.4xlarge` H100 Capacity Block in US East (Amazon Web Services, 2026a). Thus these figures answer “what would the stated or estimated accelerator-hours cost at a public contemporary rate?”, not what the original authors paid. They exclude storage, egress, idle reservation time, taxes, and discounts.

Operational emissions are estimated as energy use multiplied by the data center’s power usage effectiveness (PUE) and the grid carbon intensity (Lannelongue et al., 2021; Patterson et al., 2021):

$$\hat{E}_{\text{CO}_2\text{e}} = G P_{\text{GPU}} \text{PUE} I_{\text{grid}}. \quad (3)$$

Here $\hat{E}_{\text{CO}_2\text{e}}$ is estimated operational emissions in kg CO_2e , G is aggregate GPU-hours, P_{GPU} is assumed per-GPU power draw in kW, PUE is the dimensionless data-center power usage effectiveness, and I_{grid} is grid carbon intensity in kg $\text{CO}_2\text{e}/\text{kWh}$. We use 0.60 kW for RTX PRO 6000 runs, matching NVIDIA’s stated maximum board power (NVIDIA, 2025), and a 0.70-kW H100-class proxy for all other entries except the reported A100 run, for which we use 0.40 kW (NVIDIA, 2023; 2021). We assume PUE 1.09, Google’s reported fleet-wide value (Google Cloud, 2025), and the IEA’s 2024 global-average generation intensity of 0.445 kg $\text{CO}_2\text{e}/\text{kWh}$ (International Energy Agency, 2025). For example, one BYOD stage gives $3.2 \times 0.60 \times 1.09 \times 0.445 = 0.93$ kg CO_2e . These are location-independent GPU-only operating scenarios: they assume full rated GPU power, omit host CPU/memory/network energy and embodied hardware emissions, and do not account for renewable-energy procurement. Direct power telemetry and the datacenter’s time- and location-specific marginal carbon intensity would be required for a measured footprint.

G ADAPTER-MERGING ABLATION

Table 11: Open-generation comparison of the adapter-loaded and merged Llama checkpoints. PPL is token-weighted; Distinct- n is averaged over answers and reported as a percentage. Both use 128 generated tokens; D-1/2/3 use 32 NFE.

Representation	Parameters	PPL ₃₂ ↓	PPL ₁₂₈ ↓	D-1%↑	D-2%↑	D-3%↑
Adapter loaded	8.466B	19.381	3.753	49.5	78.8	89.9
Merged standalone	8.030B	19.283	3.720	49.8	80.7	91.8

Merging is an algebraic reparameterization rather than an additional training stage: each base projection W is replaced by $W + (\alpha/r)BA$, where A and B are the trained rank- r LoRA matrices and α is their scaling parameter. Subject to floating-point rounding, evaluating the folded projection is equivalent to evaluating the base and LoRA branches separately. The open-generation metrics are slightly better after merging, whereas the factual benchmarks show mixed task differences and a 2.0-point lower macro mean. These reruns therefore support preservation of overall behavior, but are not precise enough to attribute task-level changes to merging; a logit-level deterministic parity test would isolate numerical merge error from generation and benchmark variance.

Table 12: Factual benchmark accuracy (%) before and after merging in the matched harness; the mean is unweighted across the nine tasks.

Task	Adapter	Merged	Task	Adapter	Merged
ARC-C	82.8	82.0	MATH	22.0	22.0
GPQA	28.0	30.0	MBPP	46.0	44.0
GSM8K	65.2	58.0	MMLU	61.2	64.0
HellaSwag	73.2	68.0	MMLU-Pro	33.2	28.0
HumanEval	50.0	48.0	Mean	51.3	49.3